

Ultrafiltration

Group #1

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Chemical Engineering Lab II
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Introduction

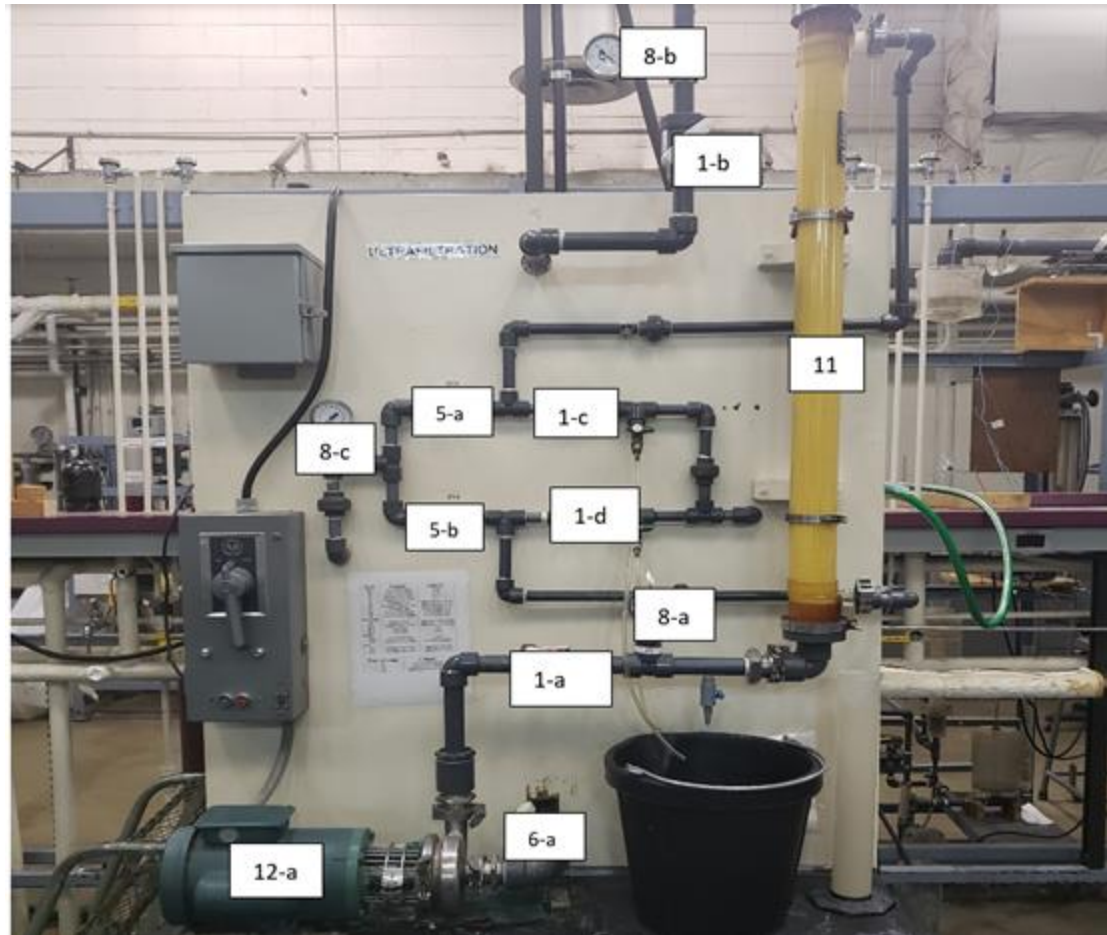
- **Purpose of lab:** Using membranes to separate the water from milk solution.
- **Technical nature:** membrane unit is a hollow fiber module.

Theory

- **Equation 1:** $v = Q_c \Delta P = (\text{steps } \dots) = k_c \ln \left(\frac{C_s}{C_1} \right)$
- **Equation 2:** $\text{C.F.} = \frac{V_{\text{original}}}{V_{\text{original}} - V_{\text{permeate}}}$
- **Equation 3:** $\% \text{Recovery} = \frac{V_{\text{permeate}}}{V_{\text{original}}} \cdot 100\%$
- **Concepts:** small pores, flux, diffusivity, and diffusion.

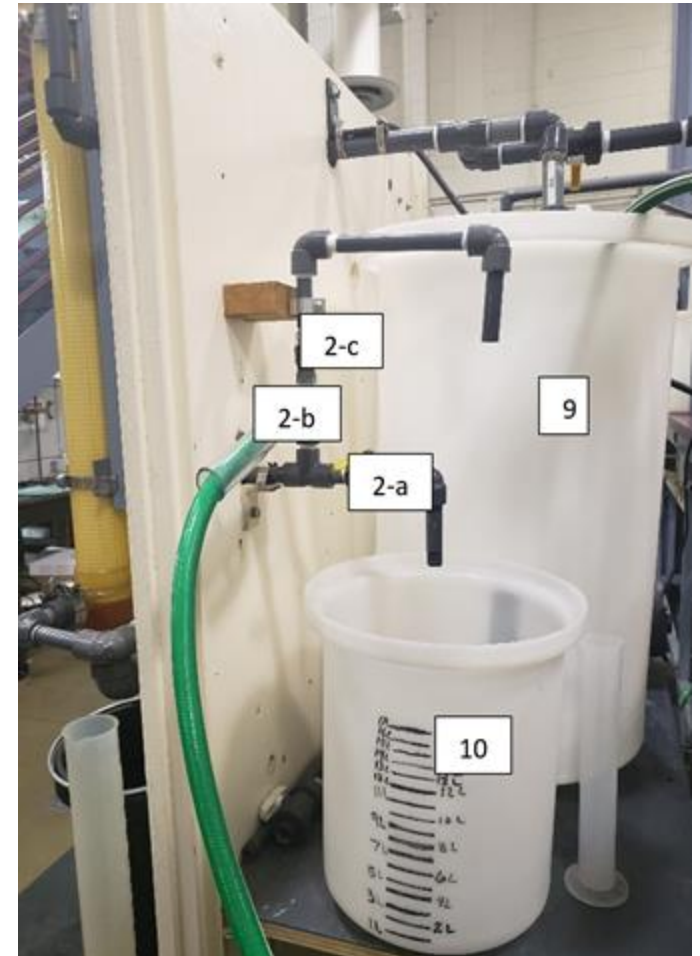
Apparatus (part 1)

- Filter (11)
- Pump (12-a)
- Pressure gauges (8-a and 8-b)



Apparatus (part 2)

- Process Tank (9)
- Permeate Tank (10)
- Recycle loop (2-b)
- Exit (2-a)



Material & Supplies

- Milk
- DI Water
- Graduated Cylinder
- Phosphoric Acid
- Sodium Hydroxide
- Stirring Rod
- Bleach
- Litmus Paper
- Goggles and Gloves

Procedure (part 1)

- **Step 1:** Do the prelab cleaning procedure (involves flushing the system with DI water, phosphoric acid, and sodium hydroxide).
- **Step 2:** Add ~20 L of DI water to the process tank with hot water. Change the outlet pressure and collect water to find relationship between ΔP and flux.

Procedure (part 2)

- **Step 3:** Drain the process tank. Add in 40 L of hot water, add in the chosen milk, and then finish adding in another 40 L of hot water (stir it up so that it is well-mixed).
- **Step 4:** Open up the exit valve while closing the recycle loop, run the system until 10% recovery is reached (~8.4 L).

Procedure (part 3)

- **Step 5:** Close exit valve, reopen recycle loop, and repeat step 2 for the part of changing ΔP and measuring flux.
- **Step 6:** Repeat step 4 until 25% recovery is reached (~ 12.6 L).
- **Step 7:** Repeat step 5.

Procedure (part 4)

- **Step 8:** Repeat step 4 until 40% recovery is reached (~12.6 L)
- **Step 9:** Repeat step 4 until 25% recovery is reached (~12.6 L).
- **Step 10:** Repeat step 5.
- **Step 11:** Clean up via back flush mode.

Experimental Challenges

- **Difficulty:** prelab cleaning takes a long time.
- **Things to watch out for 1:** Pay close attention to which valves are supposed to be open or closed.
- **Things to watch out for 2:** the yellow lever is old, the closed position is 90° away from the faucet head.

Safety

- Milk might start to smell bad after a few hours.
- Water used in the process tank is very hot.